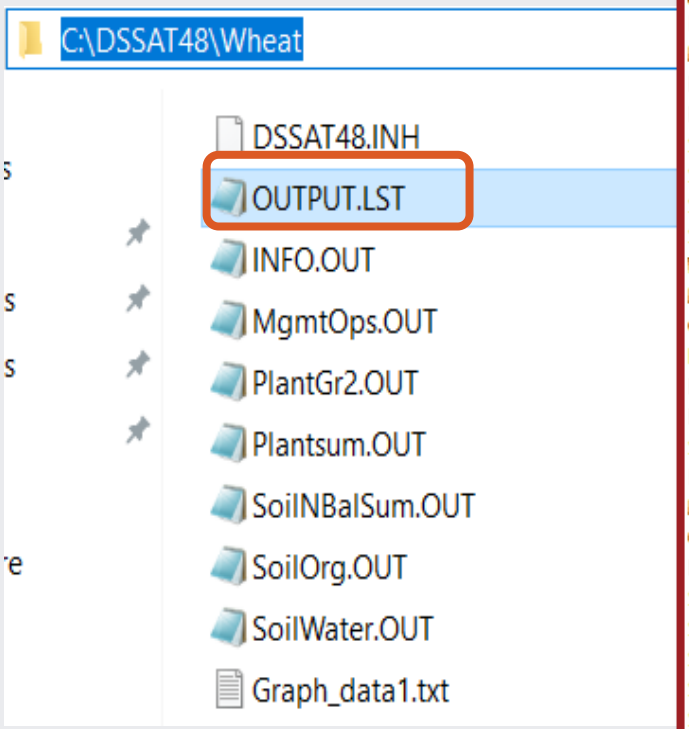
DSSAT
Outputs

DSSAT: Decision Support System for Agrotechnology Transfer

Prepared by/ Marwa Samy

List of all output files

- **C:/DDSAT48/Wheat/Output.LST**




```
*Output file list

@FILENAME          DESCRIPTION
ET.OUT             Daily soil-plant-atmosphere output file
MgmtOps.OUT        Daily management operations output file
PlantGro.OUT       Daily plant growth output file
PlantN.OUT         Daily plant nitrogen output file
SoilNi.OUT         Daily inorganic soil N output file
SoilOrg.OUT        Daily soil carbon output file
SoilTemp.OUT       Daily soil temperature output file
SoilWat.OUT        Daily soil water output file
Weather.OUT        Daily weather output file
Mulch.OUT          Daily surface mulch output file
GHG.OUT            Greenhouse gas emissions report
N2O.OUT            Soil N2O and N2 daily emissions output
PlantGr2.OUT       Wheat and barley specific output file
PlantGrf.OUT       Wheat and barley specific output file
SoilWater.OUT      Daily soil water output file by layer
Evaluate.OUT       Evaluation output file (simulated vs. measured)
MgmtEvent.OUT      Management event output file
OVERVIEW.OUT       Seasonal overview output file
Plantsum.OUT       Not currently operational
SoilCBal.OUT       Seasonal soil carbon balance
SoilCBalSum.OUT    Soil C balance summary
SoilNBalSum.OUT    Soil N balance summary
SoilNiBal.OUT      Seasonal inorganic soil N balance
SoilNoBal.OUT      Seasonal organic soil N balance
SoilWatBal.OUT     Seasonal soil water balance
Summary.OUT        Seasonal summary output file
INFO.OUT           Information output file
WARNING.OUT        Warning messages
LUN.LST            List of output files generated
```

C:\DSSAT48\DATA.CDE

Decoding simulated and field data

- *SUMMARY, *GROWTH, *NITROGEN, *WATER, *CARBON, *PESTS, *EXPERIMENTAL DATA, *USER, *WEATHER, *OTHER MODELS, *SOIL ORGANIC MATTER, *PLANT PHOSPHORUS, *SOIL POTASSIUM, *FRESH WEIGHT, *CANEGRO - South Africa, *Dynamic soil properties, *Leaves, *MOW & GRAZING, *FORAGES, *Greenhouse Gas Emissions,

```
*SUMMARY
@CDE   LABEL          DESCRIPTION.............................
ADAP   Anthesis day   Anthesis day (dap)
ADAT   Anthesis date  Anthesis date (YrDoy)
AMDY   Anth to matur  Anthesis to maturity (days)
B1DAP  Branch 1 dap   Days after planting to the first branch
B2DAP  Branch 2 dap   Days after planting to the second branch
B3DAP  Branch 3 dap   Days after planting to the third branch
B4DAP  Branch 4 dap   Days after planting to the fourth branch
B5DAP  Branch 5 dap   Days after planting to the fifth branch
B1DAT  Branch 1 date  Date of first branching (YrDoy)
B2DAT  Branch 2 date  Date of second branching  (YrDoy)
B3DAT  Branch 3 date  Date of third branching (YrDoy)
B4DAT  Branch 4 date  Date of fourth branching (YrDoy)
```

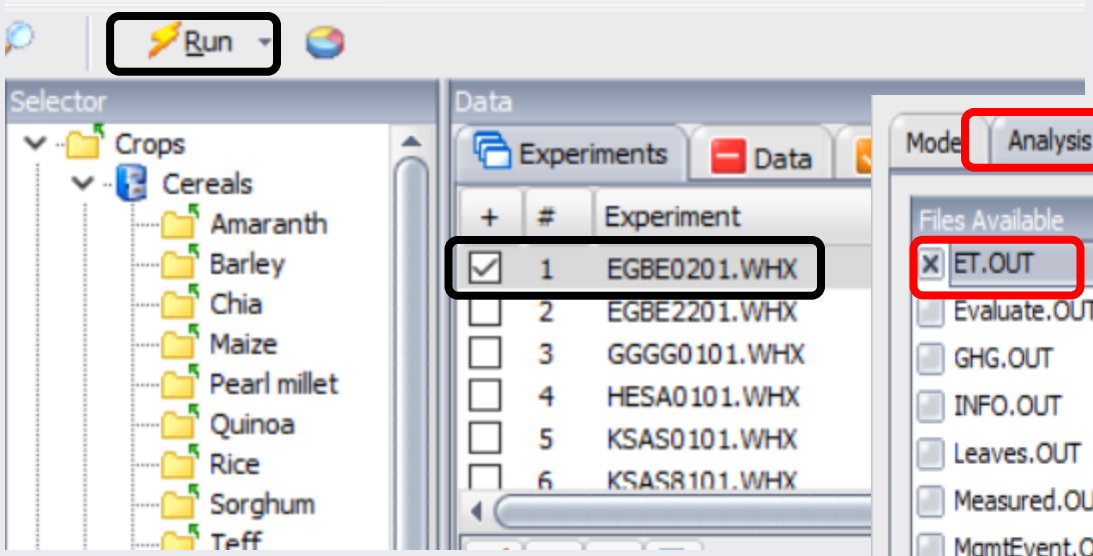
DSSAT

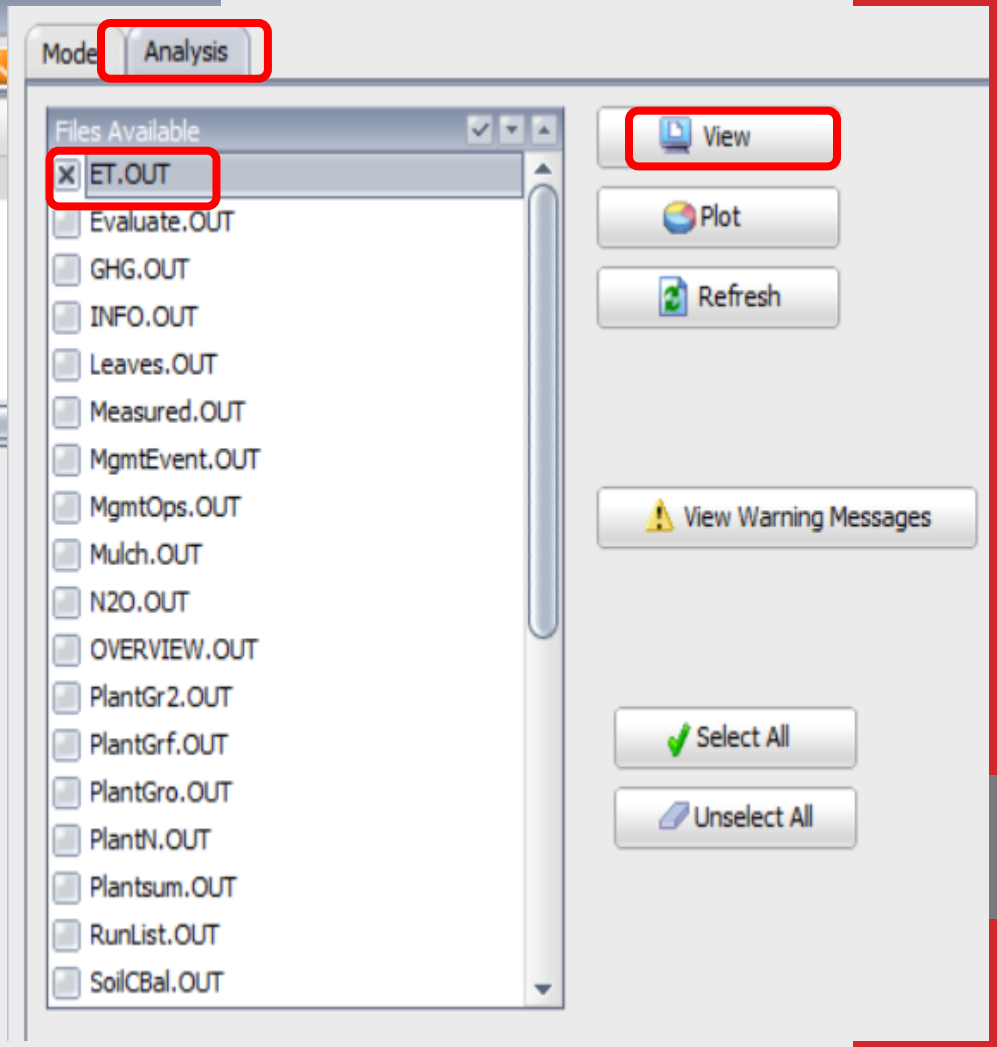
1) Summary.OUT

- **Anthesis date ADAT, Maturity date MDAT**
- **Total biomass CWAM (Kg/ha), yield HWAM**

```
.............................  DRY WEIGHT, YIELD AND Y
   ADAT     MDAT     HDAT    HYEAR  DWAP     CWAM     HWAM
2003053 2003113 2003113     2003    57    36249    16696
```

2) ET.OUT "Evapotranspiration"





1. **Mark your Experiment**
2. **Run**
3. **Mark "ET.OUT"**
4. **Analysis**
5. **view**

DSSAT

2) ET.OUT "Evapotranspiration"

File Edit Format View Help

```
*SOIL-PLANT-ATMOSPHERE MODULE OUTPUT FILE

*DSSAT Cropping System Model Ver. 4.8.5.000 -release  FEB 12, 2026 13:12:34

*RUN   1          : Sim1                        CSCER048 EGBE0201    1
 MODEL            : CSCER048 - Wheat
 EXPERIMENT       : EGBE0201 WH EGBE0201
 DATA PATH        : C:\DSSAT48\Wheat\
 TREATMENT  1     : Sim1                        CSCER048
```

@YEAR	DOY	DAS	SRAA	TMAXA	TMINA	REFA	EOAA	EOPA	EOSA	ETAA	EPAA	ESAA
2002	316	1	14.69	25.80	13.80	-99.000	3.690	0.000	3.690	3.690	0.000	3.690
2002	317	2	12.93	25.90	14.70	-99.000	3.248	0.000	3.248	2.713	0.000	2.713
2002	318	3	15.58	27.10	14.20	-99.000	3.932	0.000	3.932	2.788	0.000	2.788
2002	319	4	15.35	28.10	14.70	-99.000	3.913	0.000	3.913	1.424	0.000	1.424

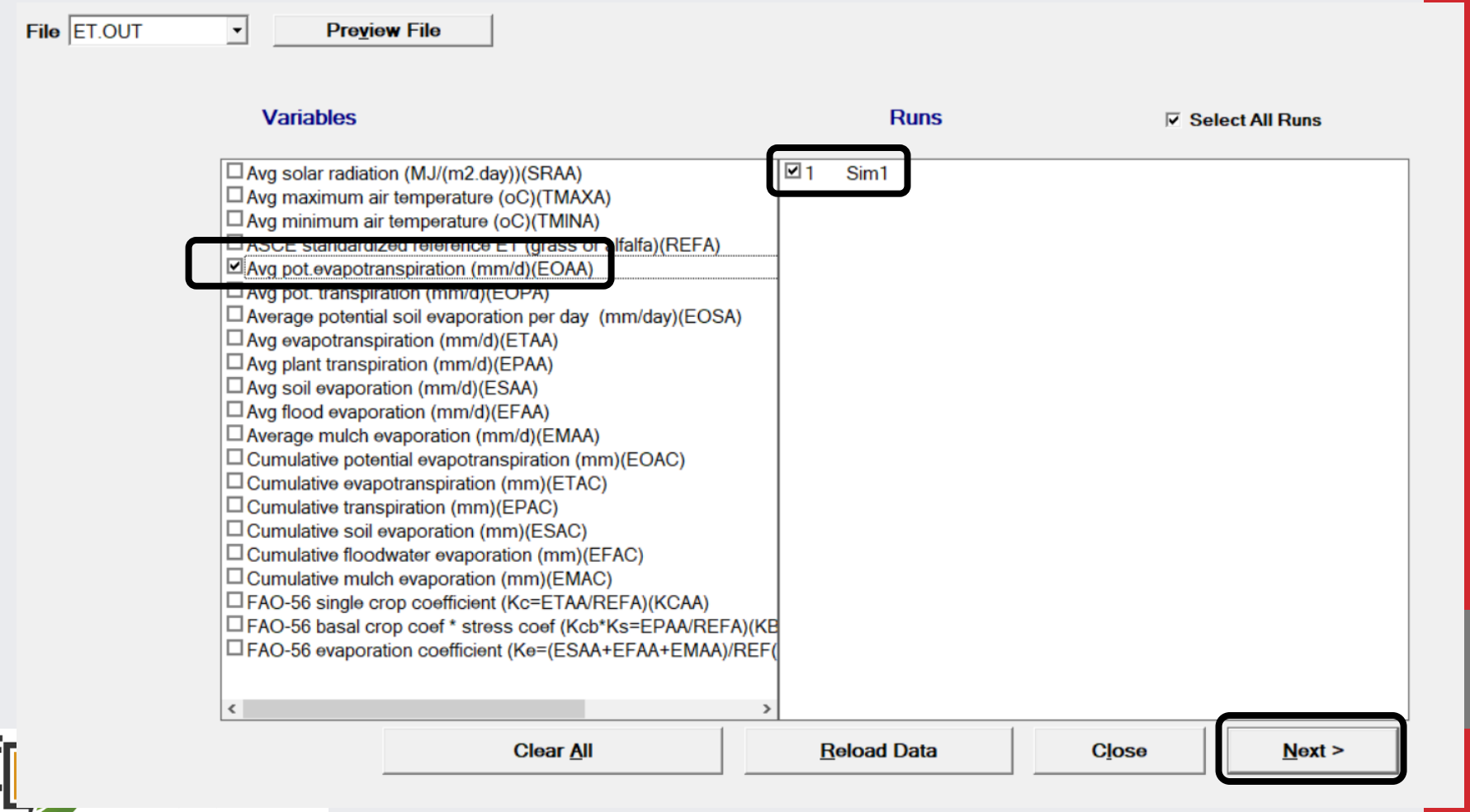
ESAA	EFAA	EMAA	EOAC	ETAC	EPAC	ESAC	EFAC	EMAC	KCAA	KBSA
3.690	0.000	0.000	3.69	3.69	0.00	3.69	0.00	0.00	-99.000	-99.000
2.713	0.000	0.000	6.94	6.40	0.00	6.40	0.00	0.00	-99.000	-99.000
2.788	0.000	0.000	10.87	9.19	0.00	9.19	0.00	0.00	-99.000	-99.000

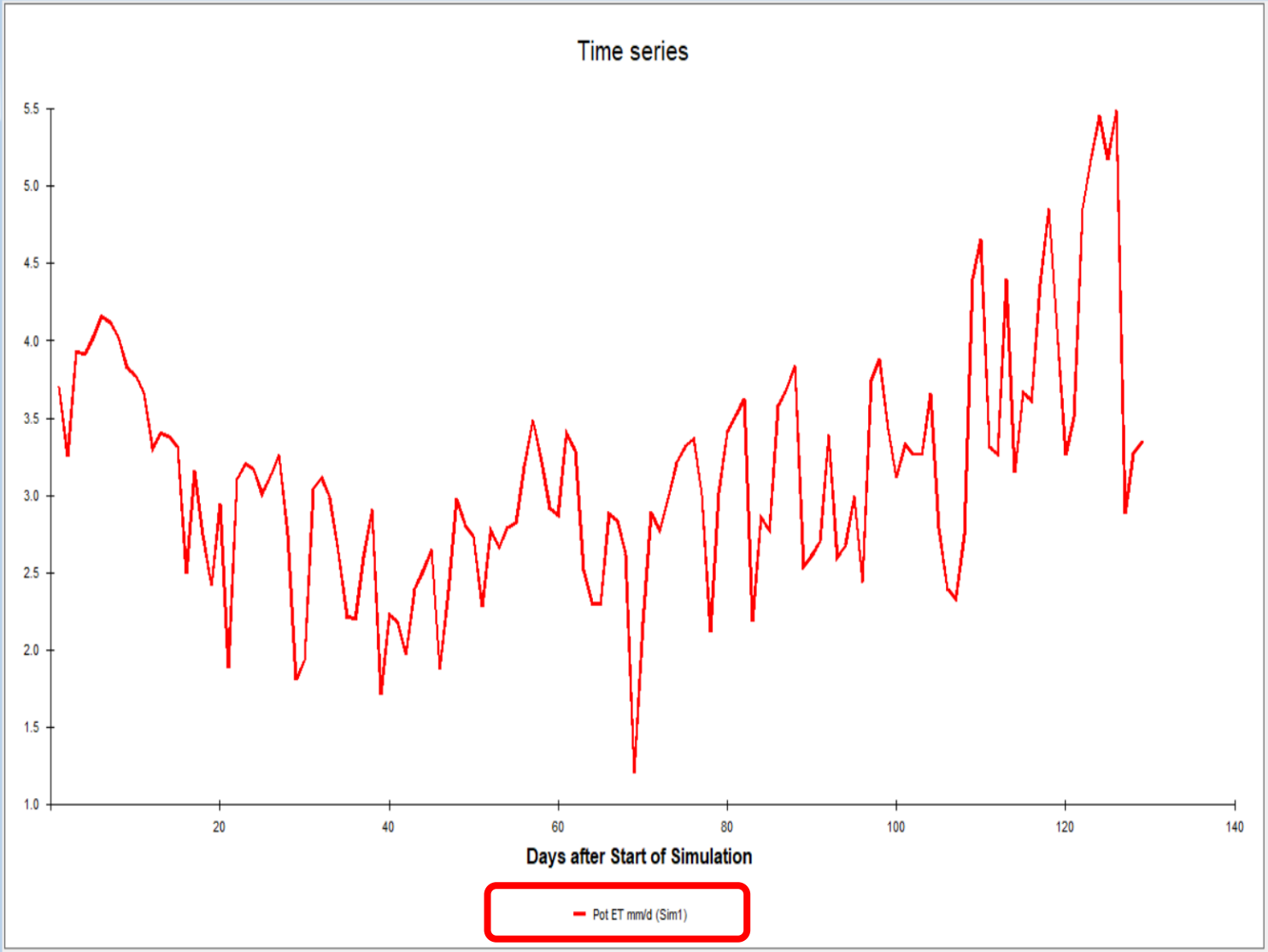
DSSAT

D. Marwa Samy

2) ET.OUT "Evapotranspiration"

Plot→ Mark "EOAA" → Sim 1 → Next





Statistic

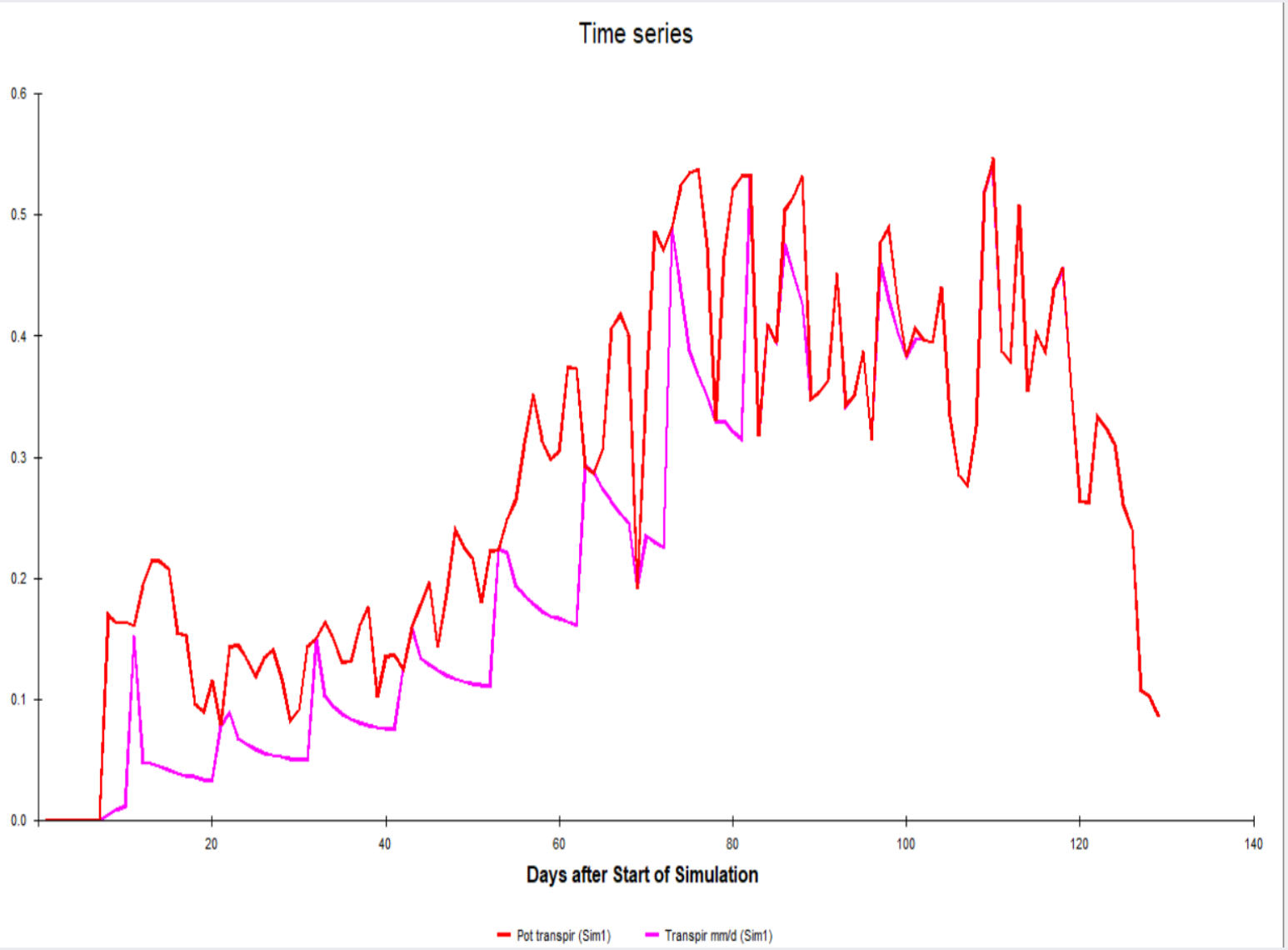
Print

Export data to text file

Export to Excel

< Back

D. Marwa Samy



Statistic

Print

Export data to text file

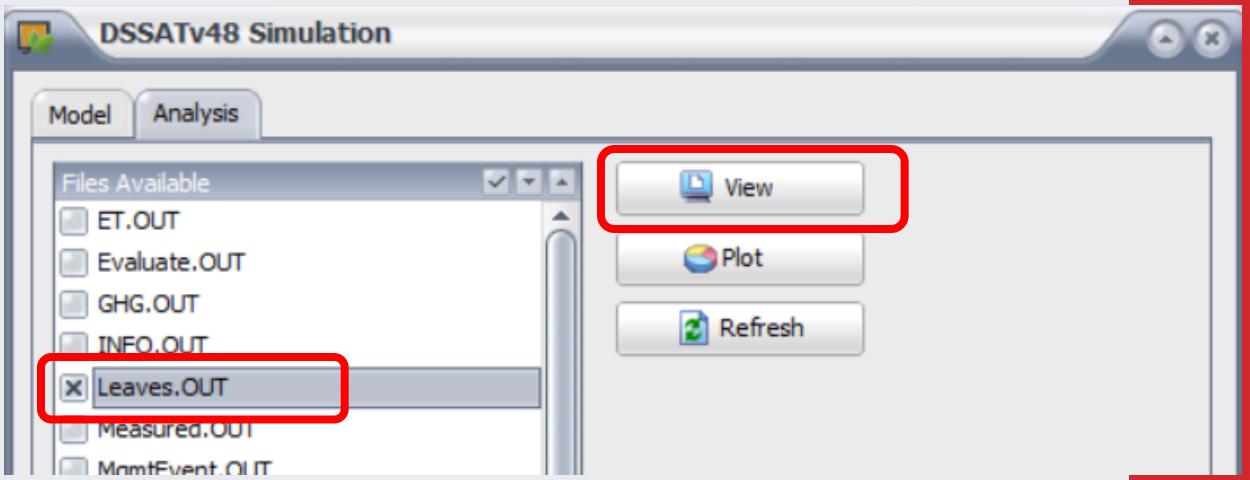
Export to Excel

< Back

D. Marwa Samy

3) Leaves.OUT

- **Run**
- **Mark "Leaves.OUT"**
- **view**



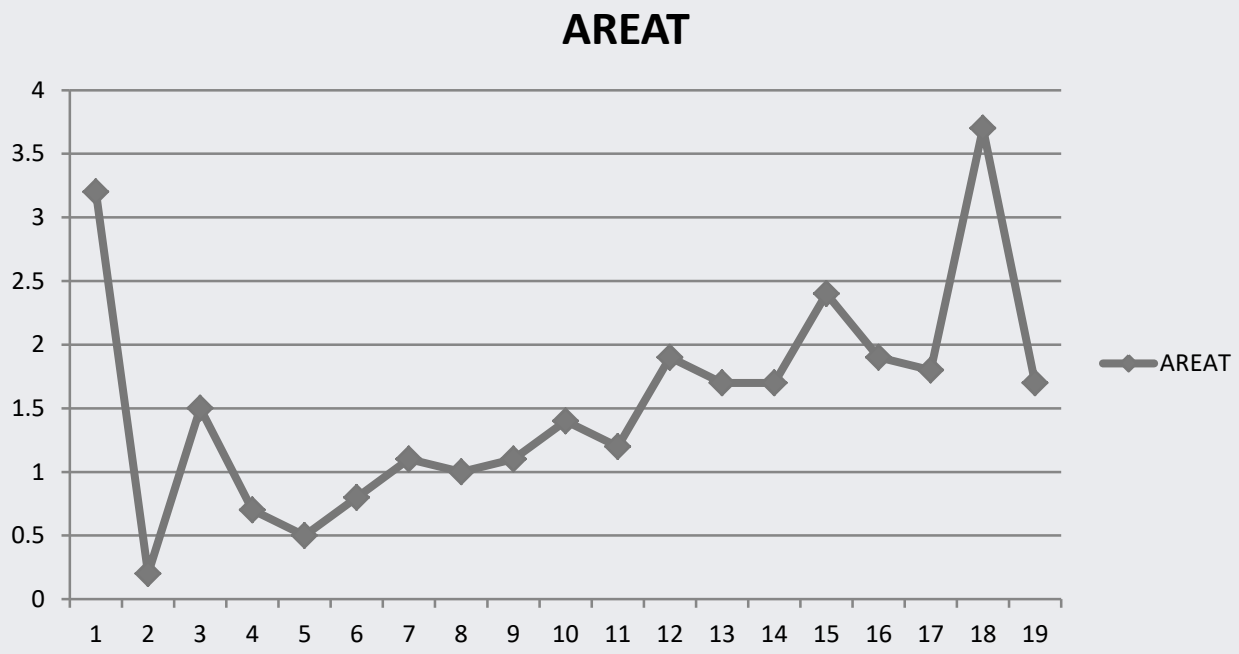
@ LNUM	AREAP	AREA1	AREAT	AREAS	T#PL	T#AL	WFLF	NFLF	AFLF	TFLF	DAYSG
1	5.0	3.2	3.2	3.2	1.0	200	0.5	0.2	0.0	0.1	2.0
2	5.5	0.2	0.2	0.2	1.0	200	1.0	0.3	0.0	0.1	1.0
3	6.1	1.5	1.5	1.5	1.0	200	0.7	0.5	0.0	0.1	3.0
4	6.7	0.7	0.7	0.7	1.0	200	0.8	0.7	0.6	0.0	3.0
5	7.3	0.5	0.5	0.5	1.0	200	0.8	0.6	0.8	0.0	3.0
6	8.1	0.8	0.8	0.8	1.2	232	0.5	0.5	0.8	0.0	3.0
7	8.9	0.9	1.1	1.1	1.2	246	0.6	0.5	0.8	0.0	4.0
8	9.7	0.8	1.0	1.0	1.2	246	0.7	0.5	0.9	0.0	3.0
9	10.7	0.8	1.1	1.1	1.6	314	0.5	0.5	0.9	0.0	4.0
10	11.8	0.9	1.4	1.4	2.0	409	0.5	0.6	0.9	0.0	4.0
11	17.7	0.5	1.2	1.2	2.3	451	0.6	0.5	1.0	0.0	4.0
12	26.5	0.7	1.9	1.9	2.8	553	0.3	0.5	1.0	0.0	4.0
13	39.8	0.6	1.7	1.7	3.1	630	0.5	0.5	1.0	0.0	4.0
14	59.7	0.5	1.7	1.7	3.3	654	0.6	0.5	1.0	0.0	4.0
15	89.5	0.7	2.4	2.4	3.9	789	0.3	0.5	1.0	0.0	4.0
16	134.3	0.6	1.9	1.9	4.1	818	0.6	0.5	1.0	0.0	4.0
17	201.4	0.4	1.8	1.8	4.3	865	0.4	0.4	1.0	0.0	3.0
18	302.2	0.9	3.7	3.7	4.1	826	0.4	0.4	1.0	0.0	5.0
19	453.2	0.5	1.7	0.2	1.8	361	0.4	0.4	1.0	0.0	3.0

```
! LNUM  = Number of leaf on main axis
! AREAP = Potential area of leaf (cm2)
! AREA1 = Area of leaf on main axis (cm2)
! AREAT = Area of leaves on all axes at leaf position (cm2)
! AREAS = Area of leaves senesced at leaf position (cm2)
! T#PL  = Tiller number/plant at leaf position
! T#AL  = Tiller number/area(m2) at leaf position
```

D. Marwa Samy

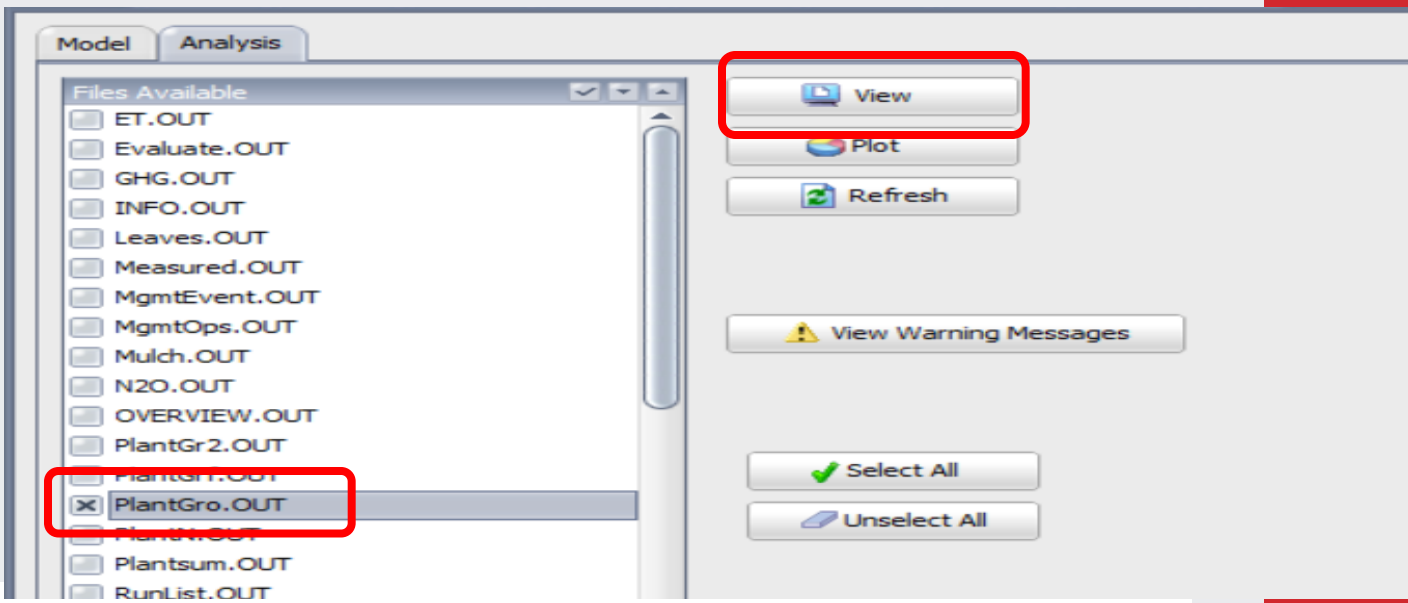
3) Leaves.OUT

- Plot → Mark “AREAT” → Next



4) PlantGro.OUT

- **Run**
- **Mark "PlantGro.OUT"**
- **View**

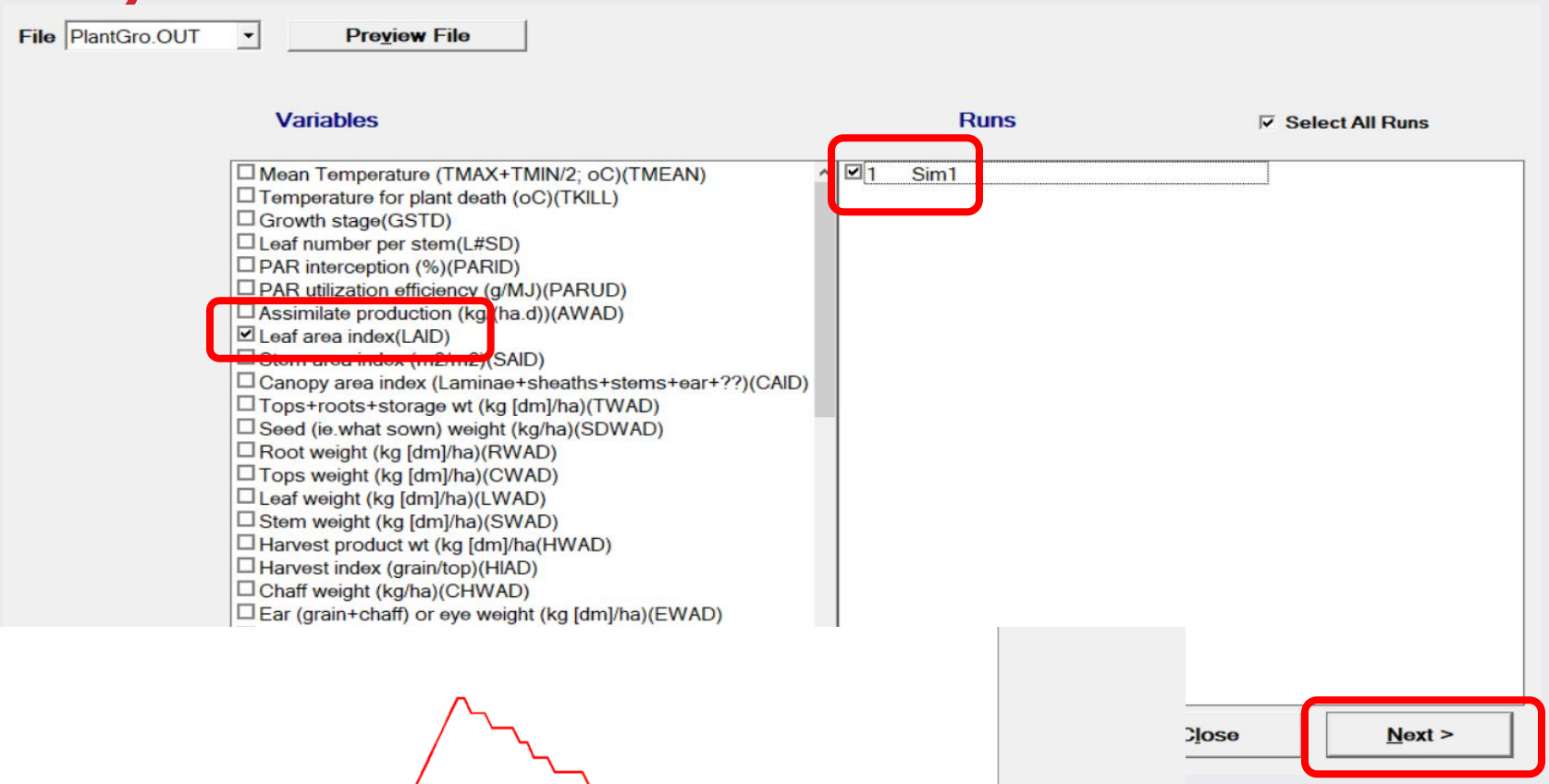


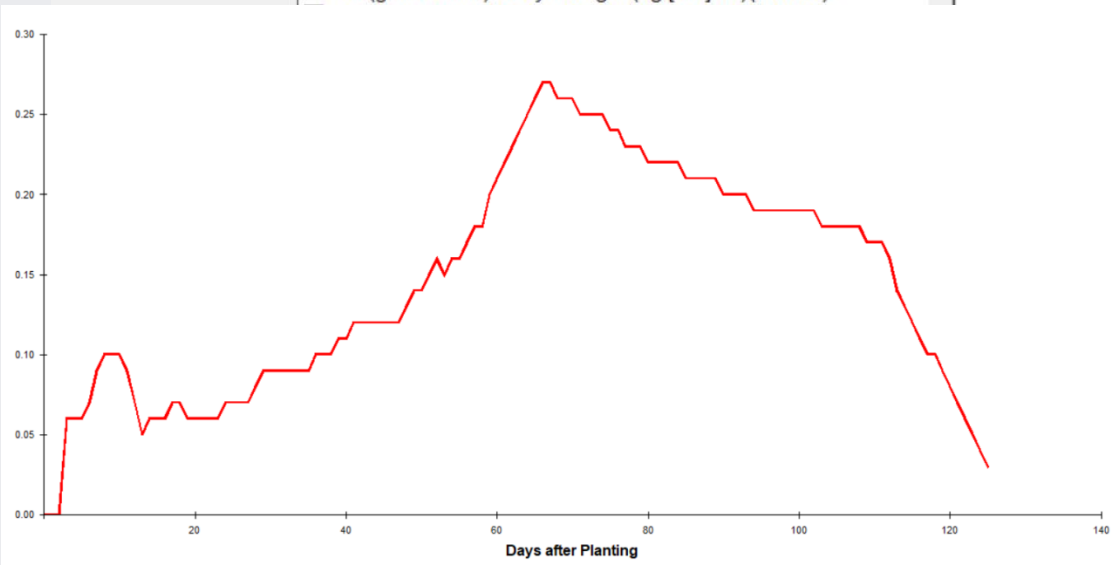

```
*DSSAT Cropping System Model Ver. 4.8.5.000 -release  FEB 12, 2026 13:12:34

*RUN   1          : Sim1                        CSCER048 EGBE0201    1
 MODEL            : CSCER048 - Wheat
 EXPERIMENT       : EGBE0201 WH EGBE0201
 DATA PATH        : C:\DSSAT48\Wheat\
 TREATMENT  1     : Sim1                        CSCER048
```

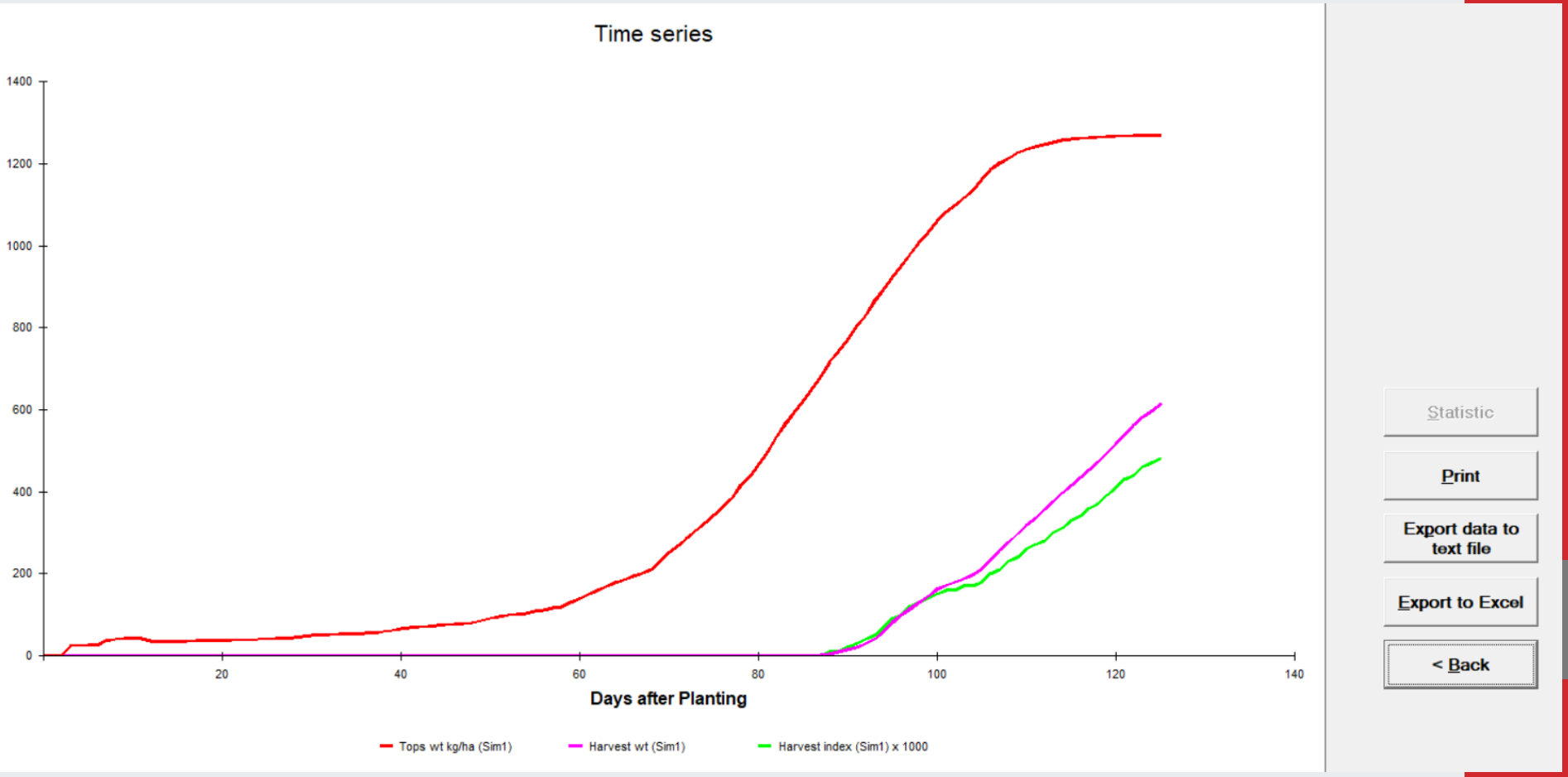
@YEAR	DOY	DAS	DAP	TMEAN	TKILL	GSTD	L#SD	PARID	PARUD	AWAD	LAID	SAID	CAID
2002	319	3	0	21.4	-6.0	0.00	0.00	0.00	0.00	0.0	0.00	0.000	0.00
2002	320	4	1	22.4	-6.0	6.77	0.00	0.00	0.00	0.0	0.00	0.000	0.00
2002	321	5	2	22.5	-6.0	10.00	0.00	0.00	0.00	0.0	0.00	0.000	0.00
2002	322	6	3	22.2	-6.0	10.74	0.74	0.05	0.00	0.0	0.06	0.005	0.07

4) PlantGro.OUT





D. Marwa Samy

4) PlantGro.OUT

- Mark Aboveground biomass in DSSAT is given by “Top weight” **CWAD**, and the economic product (yield) is the “harvest product” **HWAD**.
- To convert **kg DM/ha** “Dry Matter per hectare” → **ton/feddan**, multiply by **0.00042**